Django batch-9 Module-23 Assignment

Assignment Instructions

Develop a simple **Book Store REST API** using **Django REST Framework (DRF)**.

This assignment is designed to evaluate your understanding of the following topics covered in class:

- API & REST API
- GET, POST, PUT, PATCH & DELETE Requests
- ModelViewSet
- Token Authentication
- Filtering
- Searching
- Ordering
- Pagination
- Throttling
- Postman

Read all the instructions carefully before you start.

Requirements

1. Authentication

Implement **Token Authentication**.

Method	Endpoint	Description
POST	`/api/register/`	Register a new user
POST	`/api/login/`	Login and obtain an authentication token

Create the following APIs:Anyone can register.

- Anyone can view the list of books.
- Only authenticated users can create, update, or delete books.

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2. Book Model

Create a **Book** model with the following fields:

- Title
- Author
- Price
- Published Date

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3. Book APIs

Create the following APIs using **ModelViewSet**.

Method	Endpoint	Description
GET	`/api/books/`	Retrieve all books
POST	`/api/books/`	Create a new book
GET	`/api/books/<id>/`	Retrieve a single book
PUT	`/api/books/<id>/`	Update a book
PATCH	`/api/books/<id>/`	Partially update a book
DELETE	`/api/books/<id>/`	Delete a book

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4. Filtering, Searching & Ordering

Implement the following features for the Book API:

Filter by Author

GET /api/books/?author=J.K. Rowling

Search by Title

GET /api/books/?search=Harry

Order by Price

GET /api/books/?ordering=price

GET /api/books/?ordering=-price

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5. Pagination

Display **5 books per page**.

Example:

GET /api/books/?page=2

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6. Throttling

Configure throttling as follows:

- Anonymous Users: **20 requests per minute**
- Authenticated Users: **50 requests per minute**

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7. Postman

Test all APIs using **Postman**.

Your Postman Collection should include:

- Register API

- Login API
- Book CRUD APIs

Submission Requirements

Submit the following:

- GitHub Repository Link
- `README.md` containing:
    - Project setup instructions
    - Required packages
    - How to run the project
    - API endpoint list
- Exported Postman Collection (`optional`)